

Demand Estimation and Market Analysis: Air Fryers

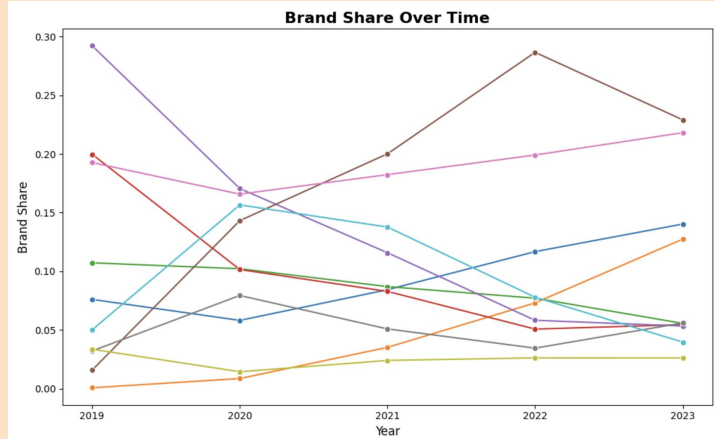
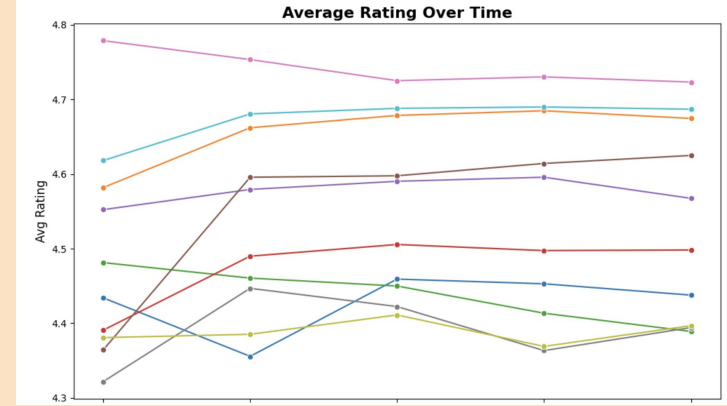
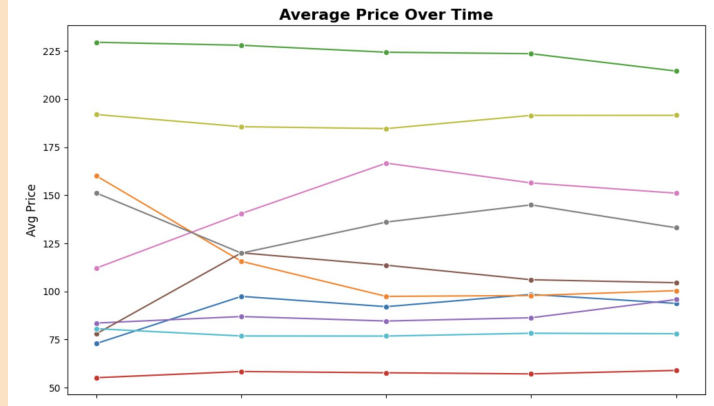
Pavi Gupta, Isabelle Lee, Lydia Lee, Kaitlin Luu

The Air Fryer Market

Among the most expensive brands in the air fryer market are Cuisinart and Oster. On the other hand, Dash and Ultrean stand at the opposite end of the spectrum, having the lowest prices. Instant Pot and Ninja have the largest shares of the market, with Ninja growing steadier than Instant Pot. Interestingly, we see that the most expensive brands in the market don't have the largest shares; the most expensive brands don't even have the highest ratings. Ninja provides a reliable product, averaging the highest rating at 4.8 stars for their air fryer. On the flipside, both Cuisinart and Oster average lower ratings at around 4.4 stars.

Generally, the air fryer market doesn't seem that stable. This is reflected by the graph's inconsistent curves, illustrating how many brands' market shares fluctuate over time. Although, there are still a few curves that stay low and stable throughout 2019-2023.

MARKET PLOTS



Legend

The Demand Model

The demand model our group created helps turn the cleaned data into a model that's usable for making predictions. The model also allows for interpretability by providing us coefficient numbers for various features that reflect:

- The relationship between price and demand
- Which product features have the highest demand
- Which brands have the highest log market shares
- Which years had the highest demands for air fryers

DEMAND MODEL

Estimated Price Coefficient

The estimated price coefficient is -0.0377. The coefficient's negative sign matters because it illustrates an inverse relationship where as price increases, demand decreases (and everything else is constant). In other words, for every 1 unit increase in price, the demand decreases by 0.0377 units. This inverse relationship highlights that the model is behaving as expected.

Most Important Product Features

The most important product features for an air fryer were a viewing window and a compact size. We're able to conclude this because the feature names "window_share" and "compact_share" had coefficients of 12.88 and 9.82 as predicted by the model. We're able to use these coefficient numbers to interpret which features are the most strongly associated with higher demands. The higher demands reflect that a viewing window and compactibility are important to consumers.

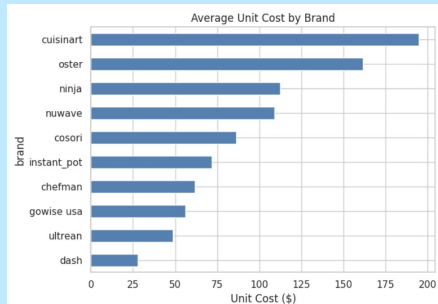
	feature	coefficient
0	avg_price	-0.037668
1	avg_rating	0.287517
2	compact_share	9.815304
3	dual_basket_share	-9.509686
4	oven_style_share	1.941774
5	rotisserie_share	-5.674054
6	window_share	12.880298
7	brand_cosori	2.551946
8	brand_cuisinart	6.422436
9	brand_dash	0.176655
10	brand_gowise usa	3.938996
11	brand_instant_pot	4.626260
12	brand_ninja	5.838705
13	brand_nuwave	3.544883
14	brand_oster	3.928074
15	brand_ultrean	0.942399
16	year_2020	0.119071
17	year_2021	0.041900
18	year_2022	-0.098860
19	year_2023	-0.003307

Coefficient Table

STRATEGY: MARKUPS AND PROFIT

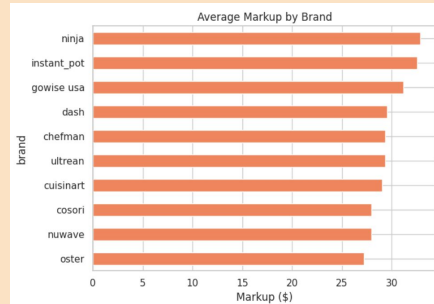
Estimated Unit Costs

Looking at unit costs, Cuisinart is the most expensive to produce, being around \$195 a unit. Oster was in second at around \$160, and Ninja was third at around \$110. On the other end, Dash had the lowest unit costs at a little over \$25, Ultrean was around \$48, and Gowise USA was around \$55. Overall, we can see that Cuisinart is the most expensive to produce, while Dash is the least expensive.



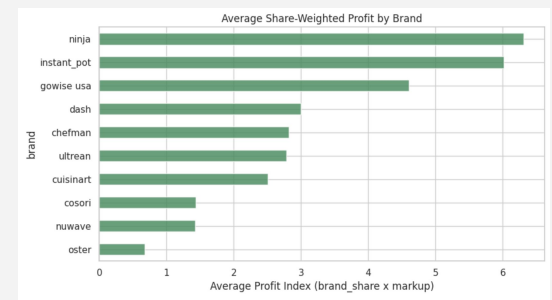
Markups

The markup cost for Ninja was the highest at around \$33. The lowest cost was Oster at around \$28. These are interesting findings, considering that Oster and Ninja were similarly ranked for estimated unit costs but at opposite ends for markups. Oster was ranked second, and Ninja was third for average price. This suggests that Oster's higher unit costs take away from their profits, while Ninja maintains better profit margins with lower unit costs.

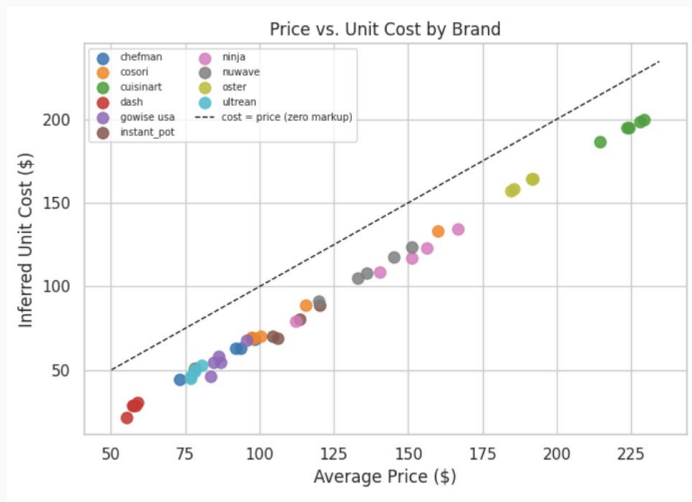


Shared-Weighted Profits

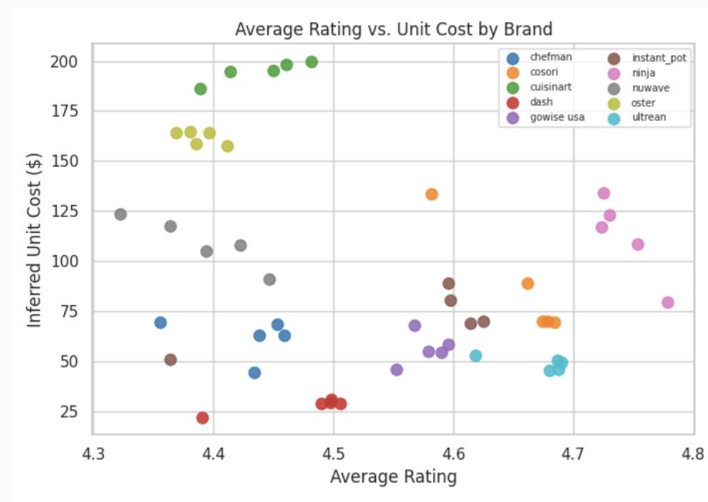
Ninja and Instant Pot have the highest share-weighted average profits, with their profit indices being 6.31 and 6.01, respectively. Gowise USA was in second with 4.61. Indices began to drop consistently with Dash, Chefman, and Ultrean, clustering between 2.8 and 3.0. Overall, this data shows how Ninja and Instant Pot are market leaders, excelling with profitability and market presence. The lower indices for the bottom brands suggest that they're more budget-friendly or have limited market reach.



Exploring Relationships



Cuisinart clusters in the top right (195–230) and Oster in the upper middle (160–200), both of which are the brands with the highest average cost. The clustering in the top right suggests that Cuisinart is produced at a high cost and sold at a high price. On the other hand, Dash clusters in the bottom left, illustrating how it's produced at a low cost and sold at a low price. Additionally, the dotted line serves as an important indicator; since we see that all dots fall below the dotted line, all the brands have positive markups.



Dash has the lowest unit costs but achieves ratings at around 4.5, which is competitive with far more expensive brands. Nuwave has high unit costs (160–200) but is only middling ratings around 4.4–4.5. Ninja sits at the high end of both cost and ratings, suggesting a genuine quality premium. Regardless, the rating vs. unit cost plot shows no clear positive trend because high cost doesn't reliably predict high ratings. Dash is a great example: a very low-cost brand earning strong consumer satisfaction. The brands that rate highest (Ninja, Cosori) aren't the most expensive, suggesting that consumer satisfaction in this market is driven more by product design and reliability than by price point.

Brands' Relative Positions in the Market

Market Leaders:

Ninja & Instant Pot

Solid Brands:

Cosori & Gowise USA

Premium but Lackluster:

Cuisinart & Oster

Most Bang for Your Buck:

Dash